

Involutory Turing Machines

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Involution f

$$\forall x \in \text{dom}(f). \quad f(f(x)) = x$$

❖ **Involution (a.k.a. involutory function)**

- ❖ A function that is own inverse
- ❖ I.e., $f(x) = y$ if and only if $f(y) = x$
- ❖ A particular kind of reversible ($\approx$ injective) function

❖ **Application of involution**

- ❖ Mathematical proofs / cryptographic systems /
Bidirectional transformation (mentioned later)

Overview

functions

$$x = y \Rightarrow f(x) = f(y)$$

injective functions

$$f(x) = f(y) \Rightarrow x = y$$

involutory functions

$$f(x) = y \Rightarrow x = f(y)$$

Computable

*Characterized by
Turing Machine (TM) etc.*

w/ function semantics [AxelsenGlück16]

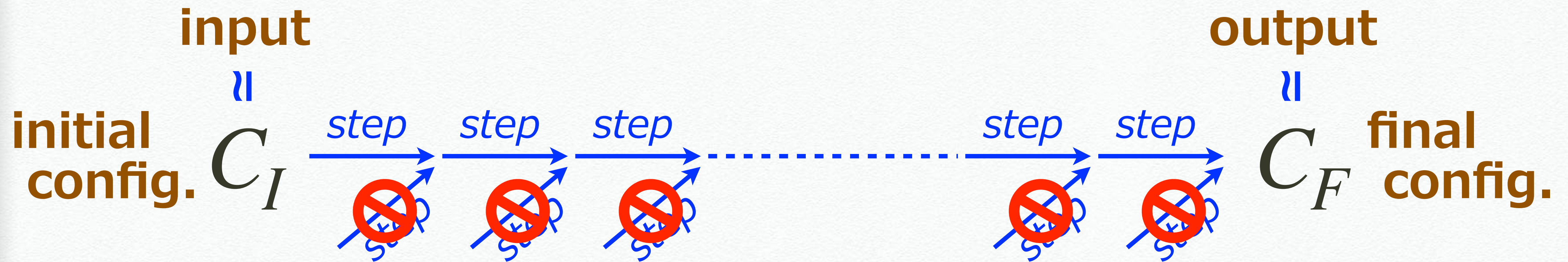
*Characterized by
Reversible TM (RTM)*

↓ This work!

*Characterized by
Involutory TM (ITM)*

Prior work on RTM [AxelsenGlück16]

❖ RTM : backward-deterministic TM



- ❖ RTM always computes an *injective* function
- ❖ Any computable *injective* function can be computed by an RTM.
- ❖ A *universal* RTM exists which simulates any RTM.

This work on ITM

- ❖ **ITM** : *somehow restricted* TM
- ❖ **ITM** always computes an *involutory* function
- ❖ Any computable *involutory* function can be computed by an **ITM**.
- ❖ A *universal* **ITM** exists which simulates any **ITM**.

Rest of This Talk

❖ **Involutory Turing Machine (ITM)**

- ❖ Definition and Semantics of TM
- ❖ Results on Reversible TM
- ❖ Definition of ITM

❖ **Properties of ITM**

- ❖ Expressiveness (Tape Reduction / Universality)

❖ **Application of ITM**

- ❖ Relationship with Bidirectional Transformation

❖ **Conclusion**

Turing Machine (TM)

$$T = (Q, \Sigma, q_I, q_F, \Delta)$$

set of states

*set of tape symbols
(except blank)*

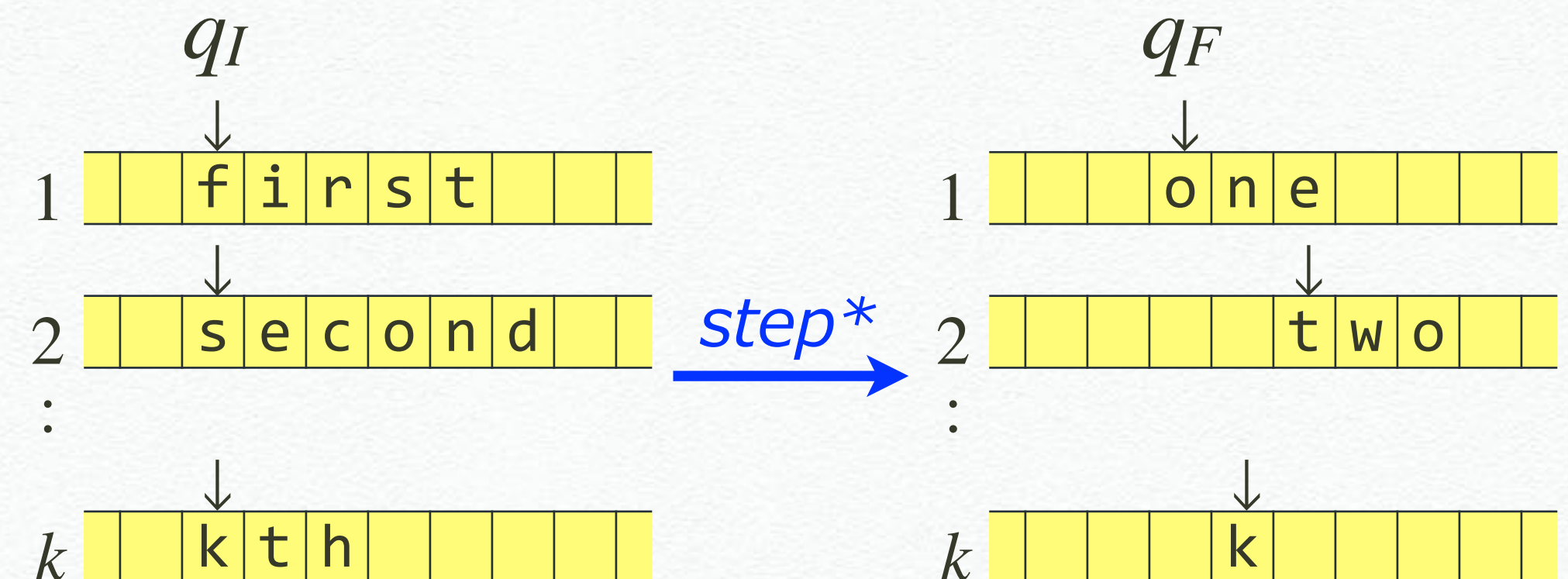
initial state in Q

final state in Q

set of transitions

❖ Working on multiple doubly-infinite tapes

- ❖ Each tape has a head.
- ❖ All left cells of the head are blank *initially* and *finally*.



Transition Rule

$$(q_1, a, q_2) \in \Delta$$

source state
in $Q - \{q_F\}$

target state
in $Q - \{q_I\}$

action

symbol

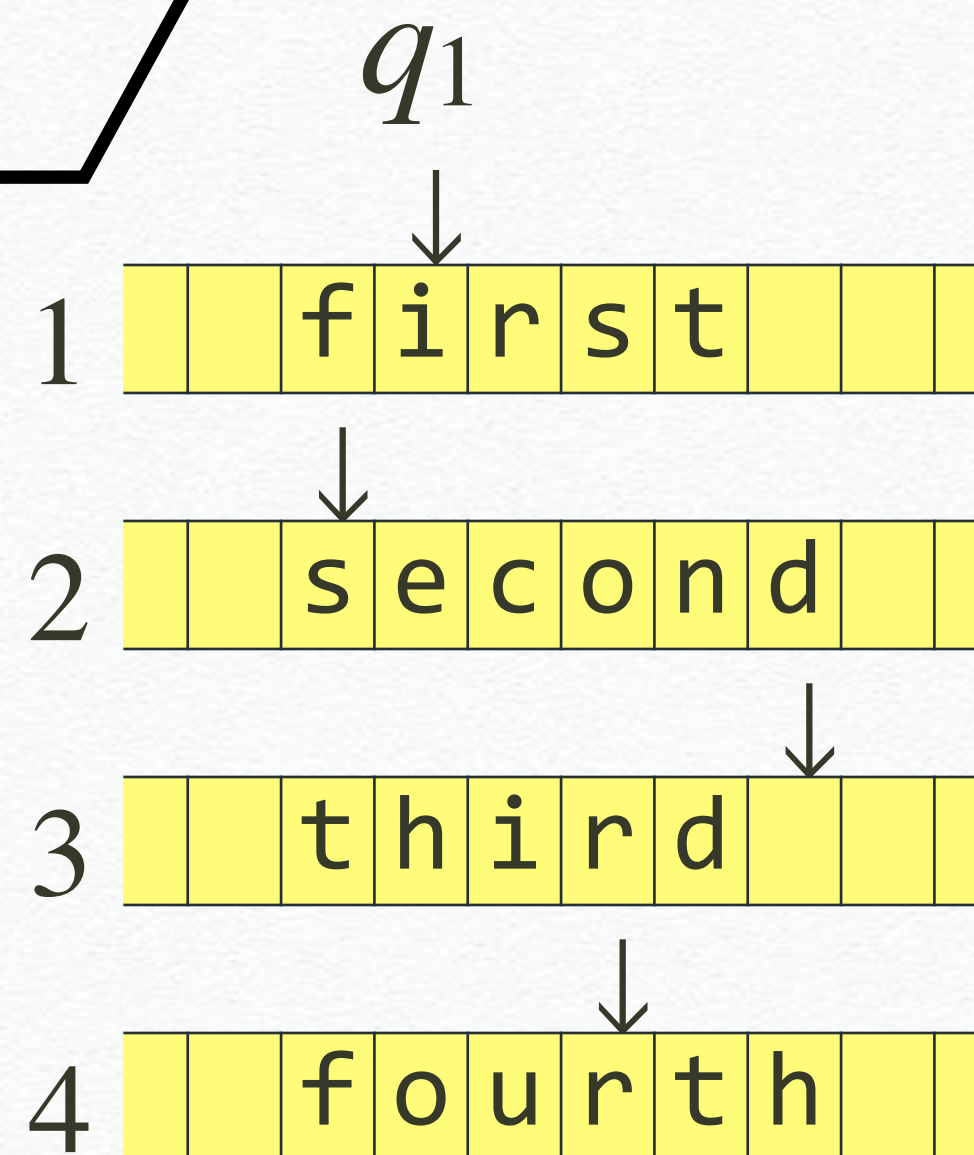
$$a \equiv s_1 \Rightarrow s_2$$

move

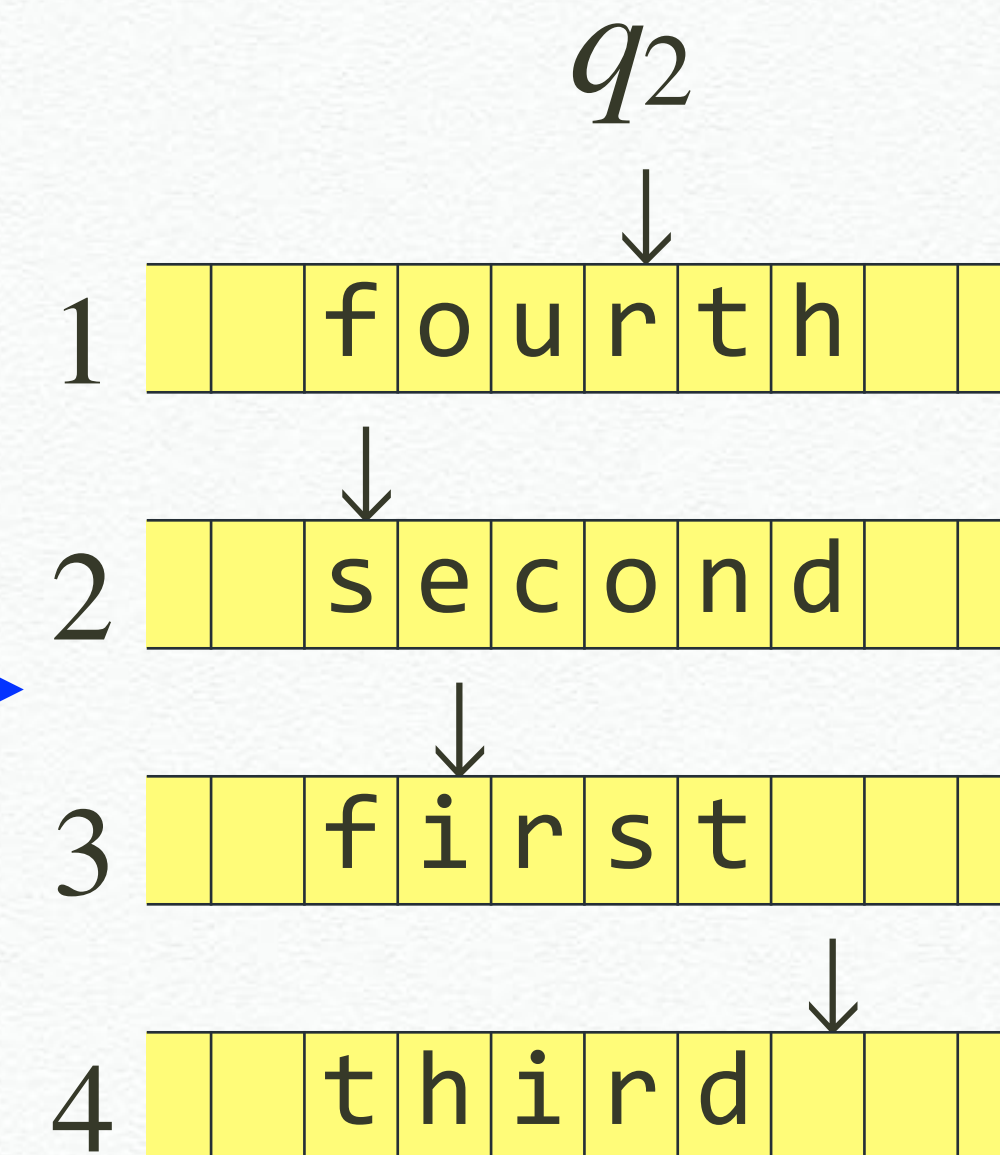
$$a \equiv \leftarrow \text{ or } \bullet \text{ or } \rightarrow$$

permutation

$$a \equiv \begin{pmatrix} 1 & 2 & \dots & k \\ i_1 & i_2 & \dots & i_k \end{pmatrix}$$



$$(q_1, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}, q_2)$$



Permute the order of the tapes
with preserving contents

Deterministic / Reversible TM

❖ Deterministic TM (DTM, or TM simply)

❖ (Locally) forward-deterministic TM

$$(\textcolor{red}{q}, a_1, -) \neq (\textcolor{red}{q}, a_2, -) \in \Delta$$

$\Rightarrow a_1$ and a_2 are symbol actions w/ different *inputs*

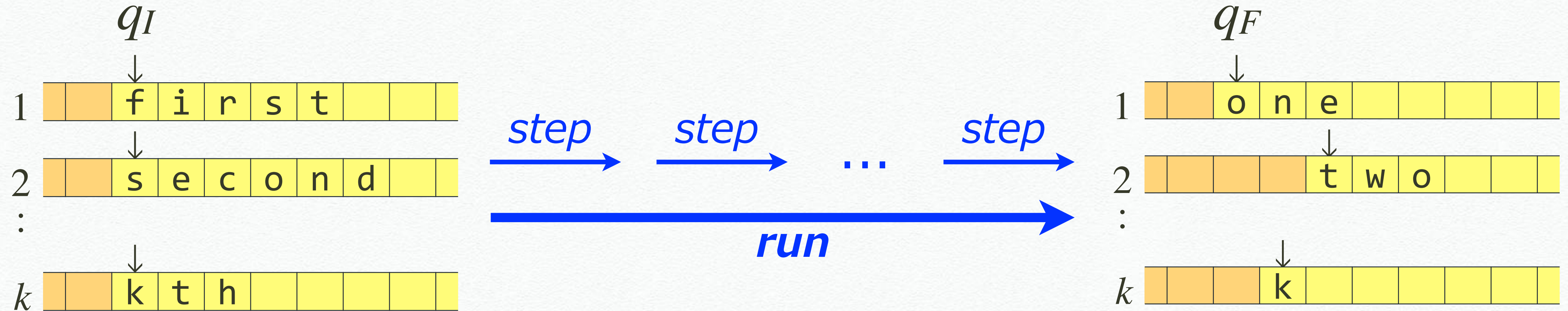
❖ Reversible TM (RTM)

❖ (Locally) forward- & backward-deterministic TM

$$(-, a_1, \textcolor{red}{q}) \neq (-, a_2, \textcolor{red}{q}) \in \Delta$$

$\Rightarrow a_1$ and a_2 are symbol actions w/ different *outputs*

Semantics of TM



$$\llbracket T \rrbracket(\text{first}, \text{second}, \dots, \text{kth}) = (\text{one}, \text{two}, \dots, k)$$

❖ **Function semantics** [AxelsenGlück16]

❖ Input/output $\approx$ initial/final configuration

Convention.

When $\llbracket T \rrbracket(x_1, \dots, x_k) = (y_1, \dots, y_k)$ implies $x_{i+1} = \dots = x_k = y_{j+1} = \dots = y_k = \varepsilon$, we may identify the function with $\llbracket T \rrbracket(x_1, \dots, x_i) = (y_1, \dots, y_j)$.

Syntactic Inverse of TM

For $T = (Q, \Sigma, q_I, q_F, \Delta)$,

$$T^{-1} \stackrel{\text{def}}{=} (Q, \Sigma, q_F, q_I, \Delta^{-1})$$

where $\Delta^{-1} = \{ (q_2, a^{-1}, q_1) \mid (q_1, a, q_2) \in \Delta \}$

$$(s_1 \Rightarrow s_2)^{-1} = s_2 \Rightarrow s_1$$

$$(\leftarrow)^{-1} = \rightarrow, \quad (\bullet)^{-1} = \bullet, \quad (\rightarrow)^{-1} = \leftarrow$$

$$\begin{pmatrix} 1 & \dots & k \\ i_1 & \dots & i_k \end{pmatrix}^{-1} = \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}$$

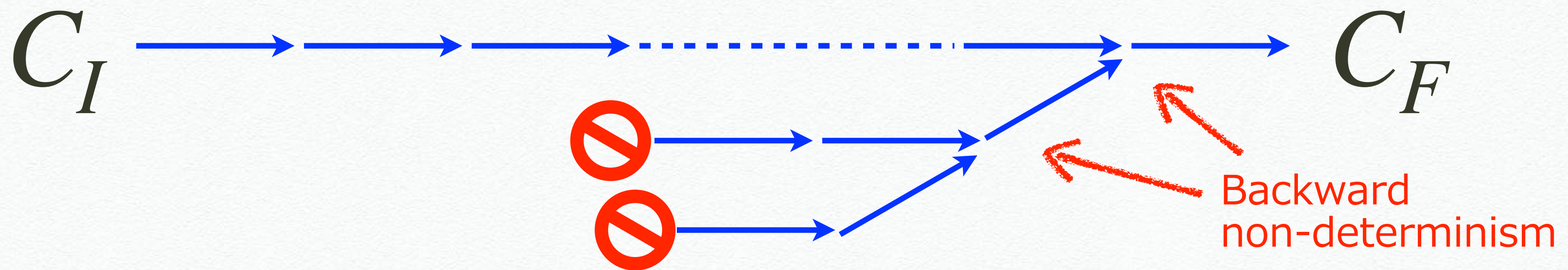
Proposition. Let T be an RTM.

T^{-1} is an RTM s.t. $\llbracket T^{-1} \rrbracket = \llbracket T \rrbracket^{-1}$.

Expressiveness of RTM [Axelsen+16]

Proposition. If the semantics of TM T is injective, then there exists an RTM T' s.t. $\llbracket T' \rrbracket = \llbracket T \rrbracket$.

❖ **Semantics of non-RTM can be injective.**



❖ Proposition implies "*an equivalent RTM always exists.*"

Cororally.

Any computable injective function can be computed by an RTM.

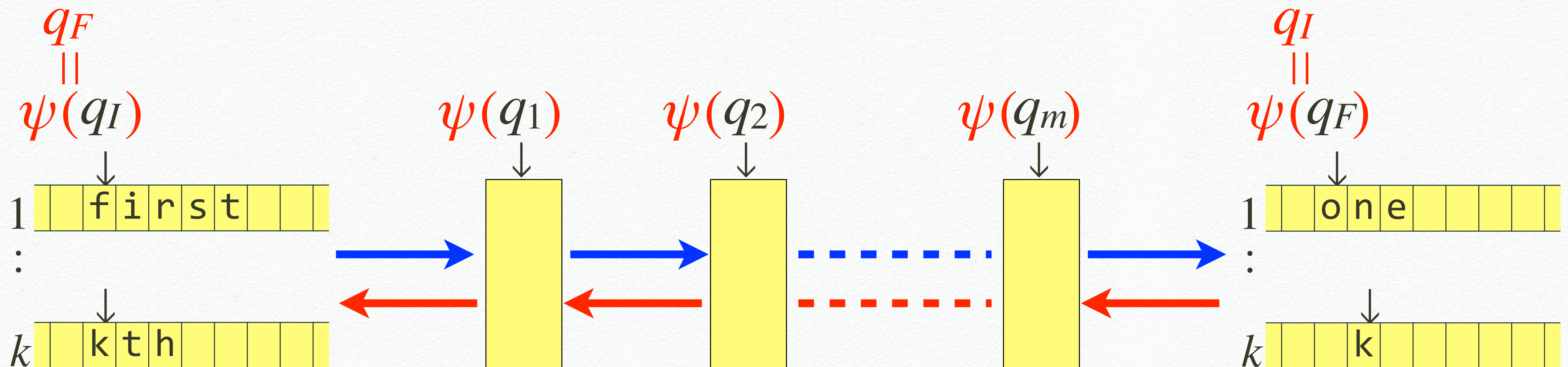
Involutory TM (ITM)

TM $T = (Q, \Sigma, q_I, q_F, \Delta)$ is *involutory* if

$\exists \psi$: involution over Q s.t.

$$\diamond \psi(q_I) = q_F$$

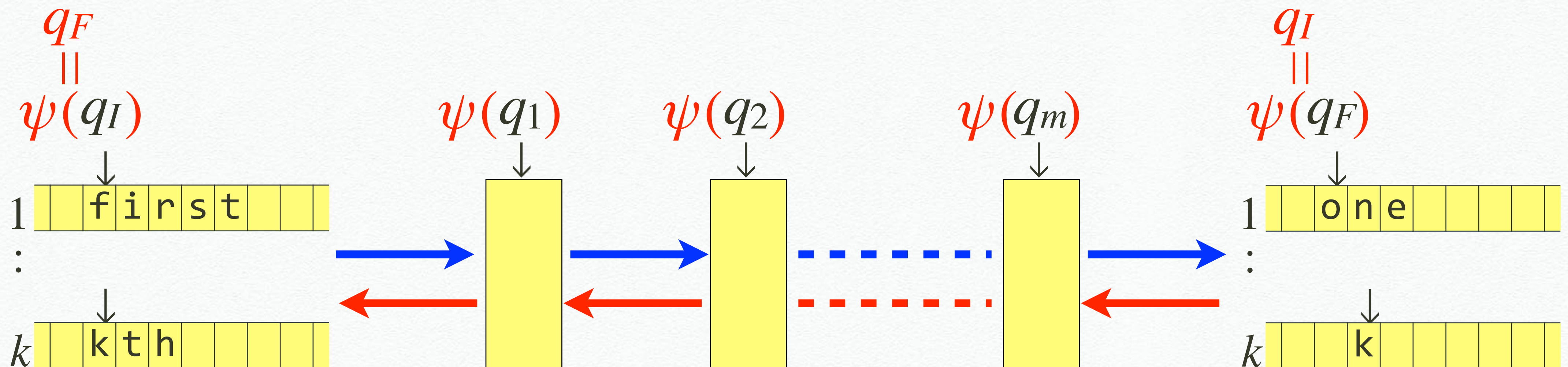
$$\diamond (q_1, a, q_2) \in \Delta \implies (\psi(q_2), a^{-1}, \psi(q_1)) \in \Delta$$



Involutory TM (ITM)

Theorem. Let T be an ITM.

Then $\llbracket T \rrbracket$ is involutory, i.e., $\llbracket T \rrbracket = \llbracket T \rrbracket^{-1}$ holds.

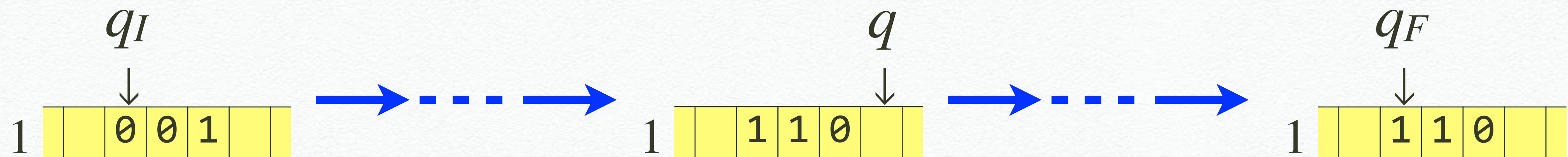


Expressiveness of ITM

Theorem. If the semantics of TM T is involutory, then there exists an ITM T' s.t. $\llbracket T' \rrbracket = \llbracket T \rrbracket$.

❖ Semantics of non-ITM can be involutory.

- ❖ Imagine a TM that computes bitwise negation by negating bits left-to-right and moving back.



- ❖ Obviously, this is not an ITM.

Cororally.

Any computable involution can be computed by an ITM.

Proof of ITM Expressiveness

Theorem. If the semantics of TM T is involutory, then there exists an ITM T' s.t. $\llbracket T' \rrbracket = \llbracket T \rrbracket$.

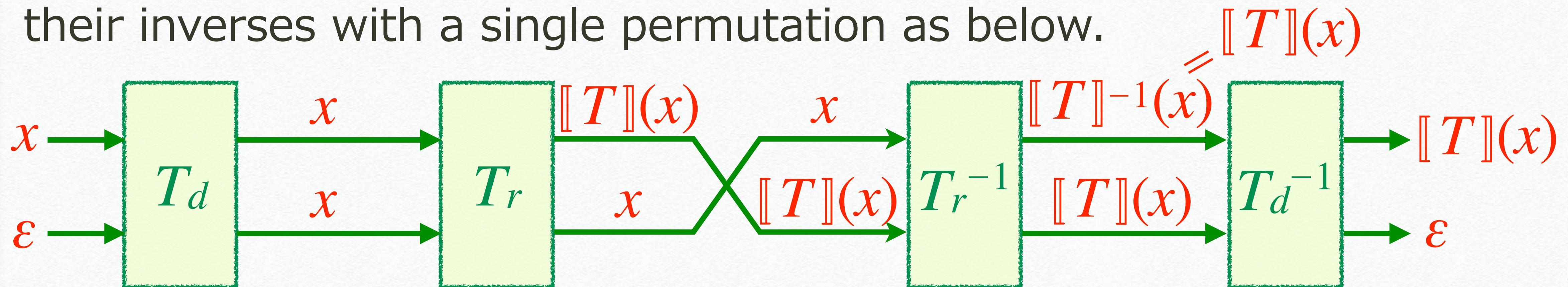
Proof sketch.

Let T be a TM s.t. $\llbracket T \rrbracket$ is involutory.

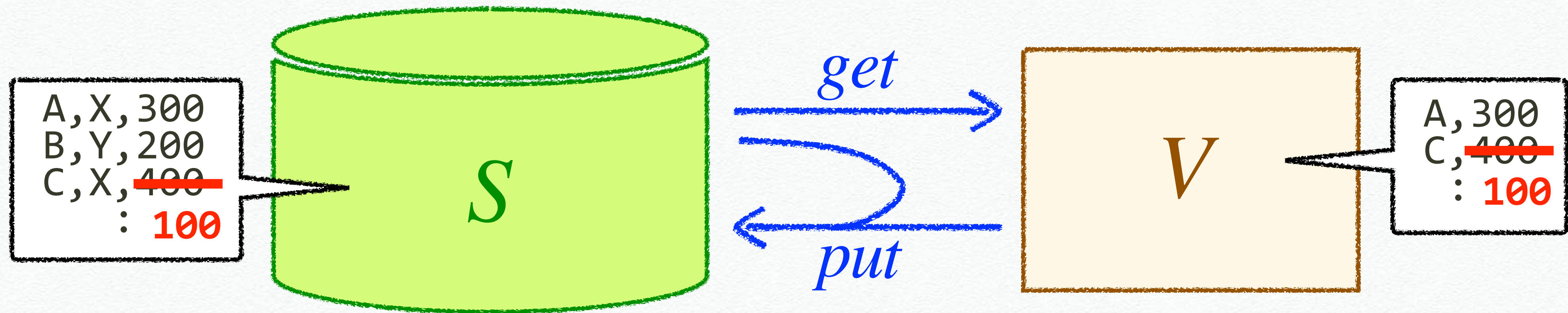
Let d and r be functions s.t. $d(x, \varepsilon) = (x, x)$ and $r(x, y) = (\llbracket T \rrbracket(x), y)$.

Due to their injectivity, we have RTMs T_d and T_r s.t. $\llbracket T_d \rrbracket = d$, $\llbracket T_r \rrbracket = r$.

An ITM T' we want is obtained by concatenating T_d , T_r and their inverses with a single permutation as below.



Applications to BX



❖ **BX: bidirectional transformation**

❖ Pair of $get: S \rightarrow V$ and $put: S \times V \rightarrow S$

❖ Characterized by $pg: S \times V \rightarrow S \times V$

such that $pg(s, v) = (put(s, v), get(s))$

❖ **Consistency forces involutoriness of pg**

❖ $pg(pg(s, v)) = (s, v)$ holds ($\bowtie$ for very-well-behaved lens)

Conclusion

- ❖ **Involutory Turing machine is presented.**
 - ❖ ITM always computes involution.
 - ❖ Any computable involution is computed by an ITM.
 - ❖ Permutation rule plays an important role for this.
 - ❖ Universal involutory Turing machine exists.
 - ❖ It can be efficiently constructed by Bennet's trick.
- ❖ **The work is motivated by my BX research.**
 - ❖ Exact computational model of BX is coming soon.